

Functions

Previous Lecture

- *Partial orders*
- *Diagrams of partial orders*
- *Functions (intro)*
 - *$f: A \rightarrow B$ is a relation R such that $(a,b) \in R$ and $(a,c) \in R$ implies $b = c$*

Domain and Codomain

- Let $f: A \rightarrow B$ be a function from A to B . Then A is called the **domain** of f , and B is called the **codomain** of f .



- If $f(a) = b$, then b is called an **image** of a , and a is called a **preimage** or **fiber** of b .
- How many images can a have via f ?
- How many preimages can b have?

Range and images

- Given a subset $C \subseteq A$ of the domain A , the **image of C** is the subset of the codomain consisting of images of elements of C :

$$f(C) = \{ b \in B \mid \exists c \in C \ f(c) = b \}$$

- The **range** of f is the image of A itself:

$$\text{range}(f) = f(A) = \{ b \in B \mid \exists a \in A \ f(a) = b \}$$

	A	B	C	D	F
Adams	1				
Chou			1		
Goodfriend		1			
Rodriguez	1				
Stevens					1

$\text{domain} = \{ \text{Adams, Chou, Goodfriend, Rodriguez, Stevens} \}$

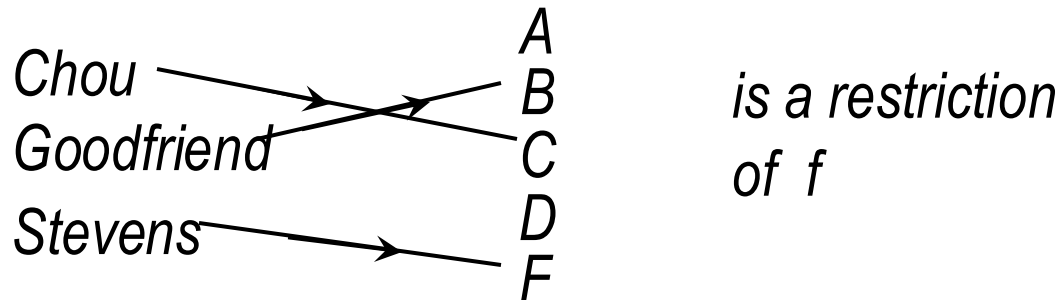
$\text{codomain} = \{ A, B, C, D, F \}$

$\text{range} = \{ A, B, C, F \}$

$f(\{\text{Adams, Chou}\}) = \{ A, C \}$

Restrictions of functions

- Let $f: A \rightarrow B$ be a function and $C \subseteq A$. A function $f|_C : C \rightarrow B$ is called a **restriction** of f to C if $f|_C(a) = f(a)$ for all $a \in C$
- Example: Let $C = \{\text{Chou}, \text{Goodfriend}, \text{Stevens}\}$. Then



- Example:
 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function $f(x) = x^2$
 Then $f|_{\mathbb{Z}} : \mathbb{Z} \rightarrow \mathbb{R}$ is the function $f(x) = x^2$ on integers

Extensions of functions

● Let $C \subseteq A$ and $f: C \rightarrow B$. Any function $g: A \rightarrow B$ such that $g(a) = f(a)$ for all $a \in C$ is called an **extension** of f .

● Example: Consider $f: \mathbb{Z} \rightarrow \mathbb{Z}$ is defined as follows:

$$f(a) = a$$

Let g be the **floor function**:

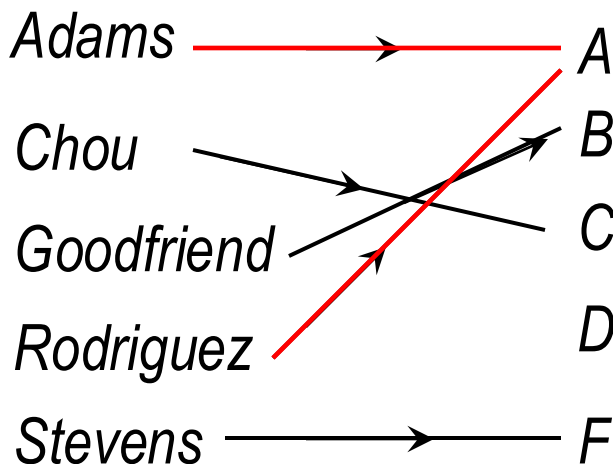
$$g(x) = \lfloor x \rfloor = \text{the greatest integer less than or equal to } x$$

Clearly, $g: \mathbb{R} \rightarrow \mathbb{Z}$, and $g(a) = a = f(a)$ for any integer a .

Thus, g is an extension of f

One-to-One Functions

- A function f is said to be **one-to-one**, or **injective**, if and only if $f(a) = f(b)$ implies $a = b$ for all a, b in the domain.
- Contrapositive is if $a \neq b$ then $f(a) \neq f(b)$ – i.e., no two distinct elements have the same image.
- Is this function injective?



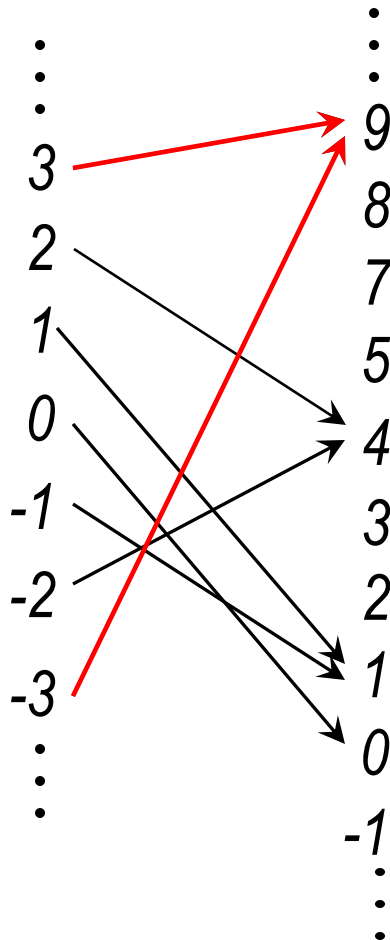
No!



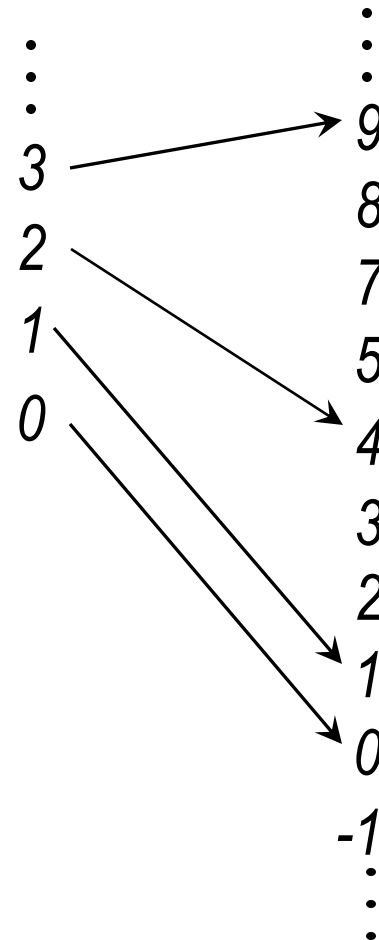
One-to-One Functions (cntd)

● Let's consider the function $f(x) = x^2$ on $\mathbb{Z}$

Is it injective?



No!



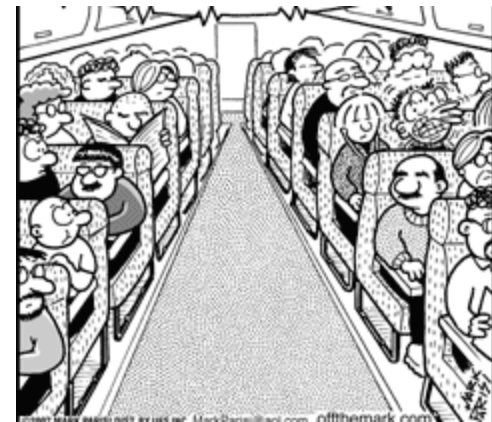
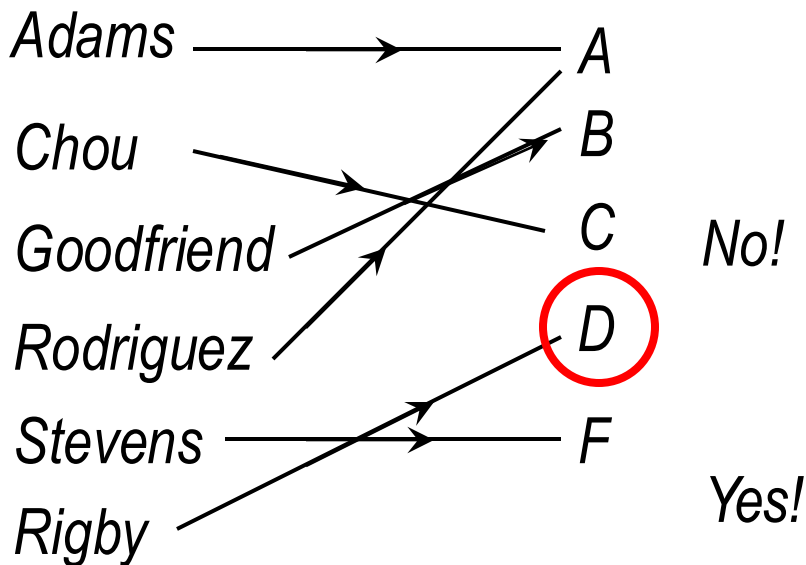
Yes!
on $\mathbb{N}$

Onto Functions

- A function f from A to B is called **onto**, or **surjective**, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$.
- Symbolically: $\forall b \exists a (f(a) = b)$
- A function is called a **surjection** if it is onto.
- Examples:
 - $f(x) = x + 1$?
 - $f(x) = x^2$ on $\mathbb{Z}$?
 - $f(x) = x^2$ on $\mathbb{R}^+$?

Onto Functions (cntd)

● More examples

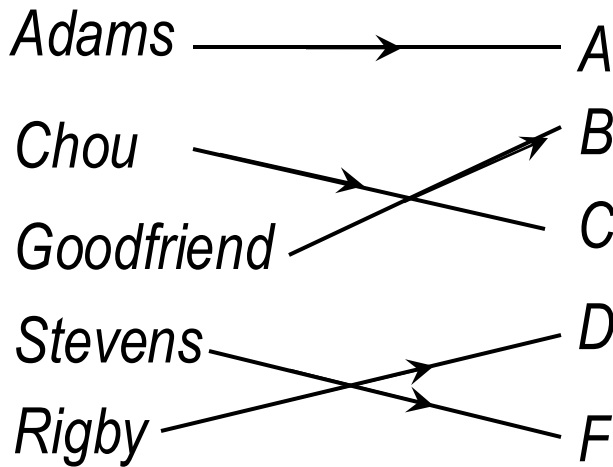


This mapping was onto!

● $f(a) = b$ if b is the father of a

Bijections

- A function f is a **one-to-one correspondence**, or a **bijection**, if it is both one-to-one and onto.



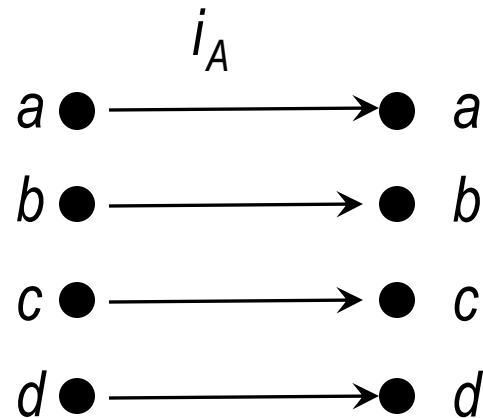
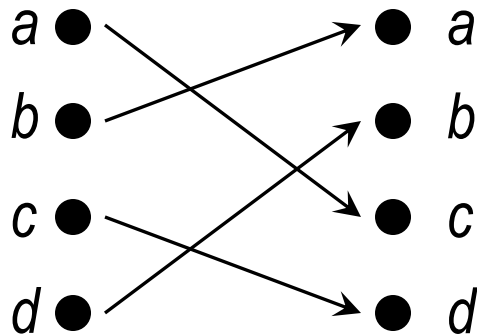
- If there is a bijection from a set A to a set B , then these sets in a certain sense are equal or identical.

Bijections (cntd)

● Numerical functions:

- $f(x) = x + 1$ is a bijection on $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$, but not on $\mathbb{N}$
- $f(x) = x^2$ is a bijection on $\mathbb{R}^+$, but is not on any other numerical set

● A bijection from a set A to the same set A is called a *permutation* of A

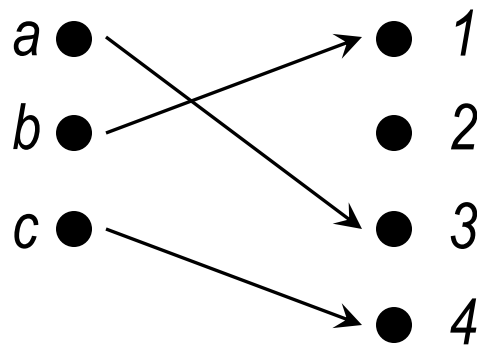


● The *identity function* on a set A is the function $i_A: A \rightarrow A$, where

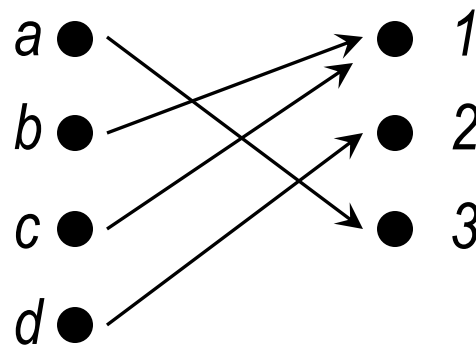
$$i_A(x) = x$$

Functions and Properties

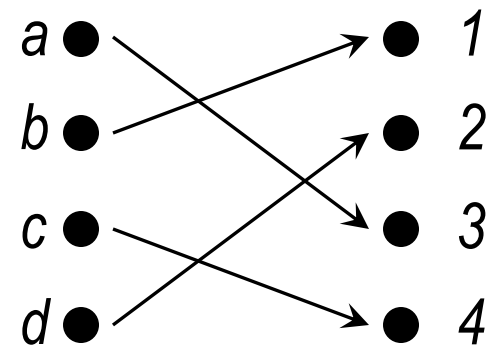
● Examples of different types of correspondences



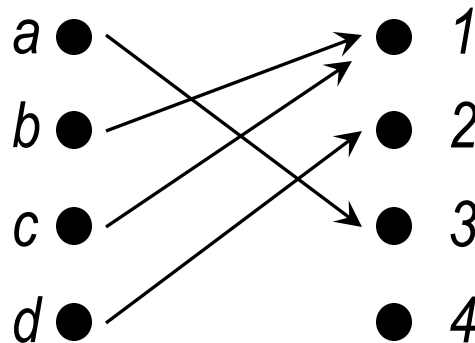
one-to-one, not onto



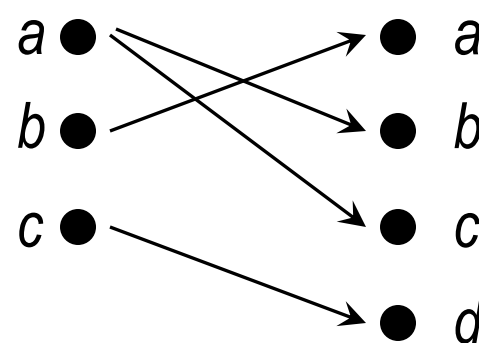
onto, not one-to-one



one-to-one and onto



neither one-to-one nor onto

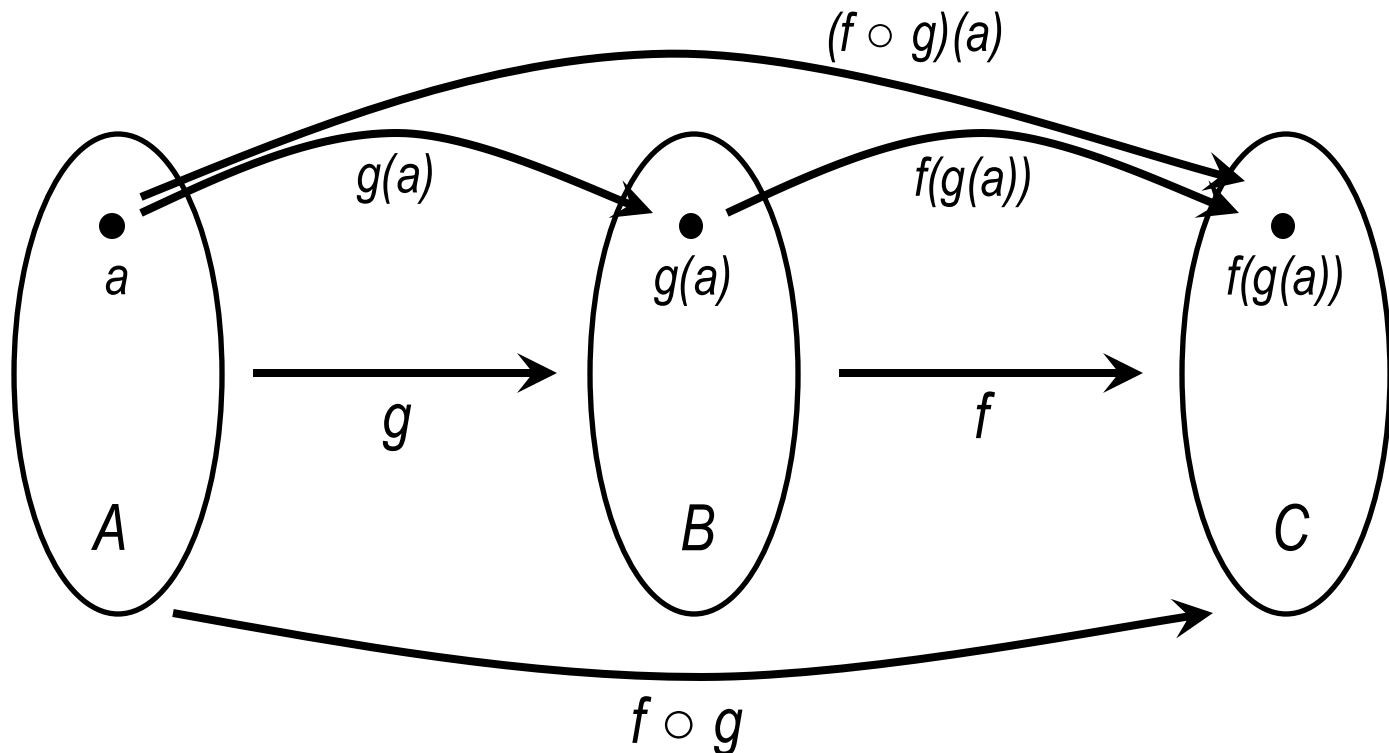


not a function

Composition of Functions

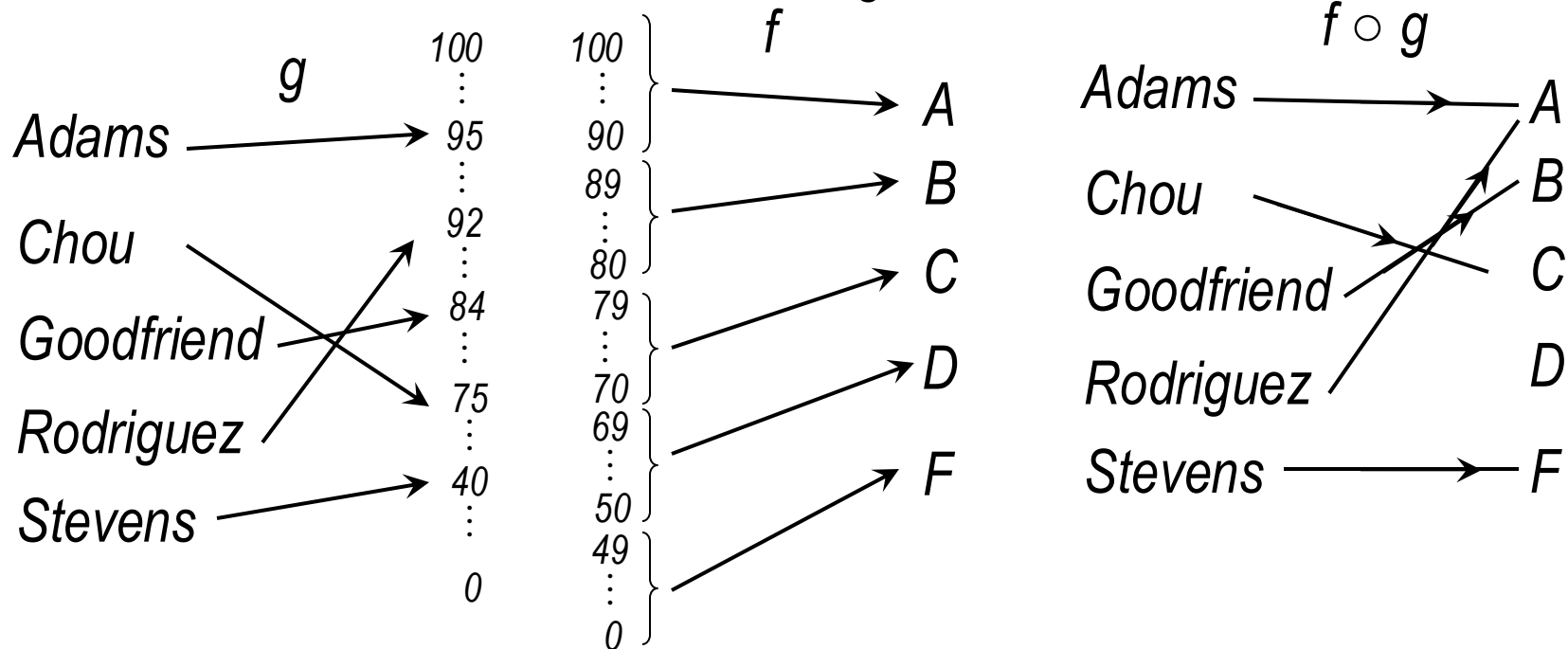
- Let g be a function from A to B and let f be a function from B to C . The **composition** of the functions f and g , denoted by $f \circ g$, is the function from A to C defined by

$$(f \circ g)(a) = f(g(a))$$



Composition of Functions (cntd)

- Suppose that the students first get numerical grades from 0 to 100 that are later converted into letter grade.



- Let $f(a) = b$ mean ' b is the father of a '.
What is $f \circ f$?

Composition of Numerical Functions

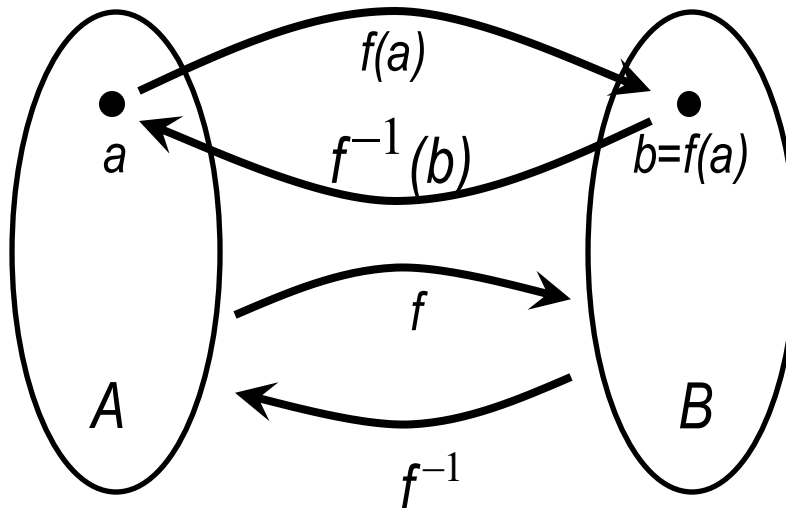
● Let $g(x) = x^2$ and $f(x) = x + 1$. Then

$$(f \circ g)(x) = f(g(x)) = g(x) + 1 = x^2 + 1$$

● Thus, to find the composition of numerical functions f and g given by formulas we have to substitute $g(x)$ instead of x in $f(x)$.

Inverse Functions

- Let f be a bijection from the set A to the set B .
- The **inverse** of f , denoted f^{-1} , is the function that assigns to an element $b \in B$ the unique preimage $a \in A$ such that $f(a) = b$.
- Logically, $f^{-1}(b) = a$ if and only if $f(a) = b$



Note!
 f^{-1} does not mean $\frac{1}{f(x)}$

$$f \circ f^{-1} = i_B$$

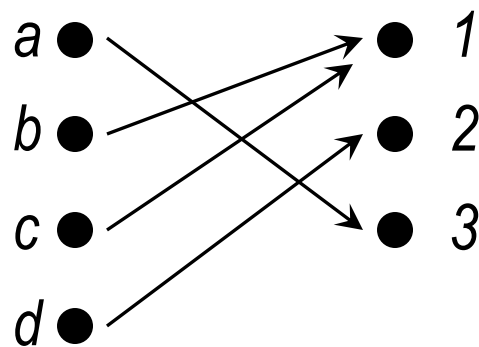
$$f^{-1} \circ f = i_A$$

Inverse Functions (cntd)

- If a function f is not a bijection, the inverse function does not exist.

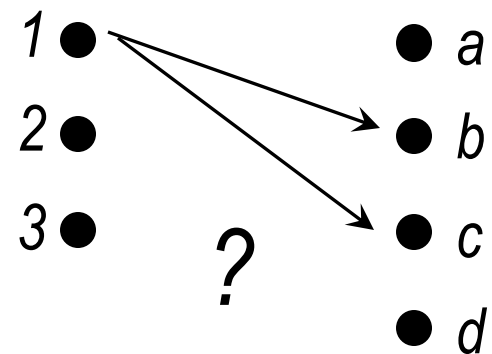
Why?

- If f is not a bijection, it is either not one-to-one, or not onto



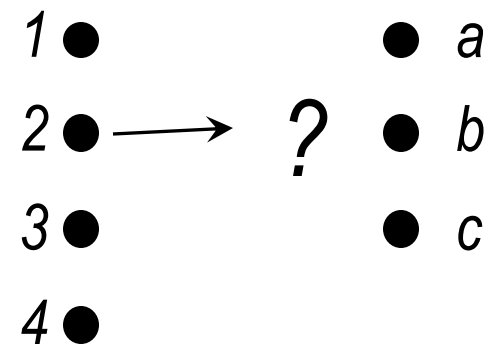
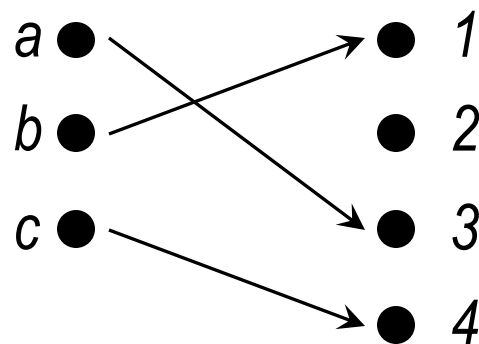
onto, not one-to-one

$$f^{-1}(1) = ?$$



one-to-one, not onto

$$f^{-1}(2) = ?$$



Practice

Exercises from the Book:

7th edition: 2, 3, 5, 10, 11, 20a, 33a, 36 (page 152 – 154)

8th edition: 2, 3, 5, 10, 11, 20a, 33a, 38 (page 161 – 163)

- *Show that the relation R on sets defined as*

*$(A,B) \in R$ if and only if there exists a bijection $f:A \rightarrow B$
is an equivalence relation*